

Tanveer Singh

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EDUCATION

University of California, Santa Barbara (UCSB)

Goleta, CA

Bachelor of Science (B.S.), Statistics and Data Science

June 2026

- Relevant Coursework: Statistical Machine Learning, Bayesian Data Analysis, Big Data Analytics, Stochastic Processes, Time Series

SKILLS

Languages: Python, R, SQL

Data Analysis: Data Visualization, Data Validation, Exploratory Data Analysis (EDA)

ML/Stats: Classification, Regression, Random Forest, XGBoost, Elastic Net, Convolutional Neural Networks (CNNs), NLP, Time Series Forecasting, Feature Engineering

Tools: Power BI, Excel, scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib, Seaborn, tidymodels, tidyverse, tidytext, ggplot2, GCP, AWS EC2, Jupyter, Docker

TECHNICAL PROJECTS

Daily Max Temperature Prediction (NOAA GSOD) | Python · BigQuery · scikit-learn June 2026

- Queried and joined NOAA GSOD 2023 station-day records with station metadata using SQL in Google BigQuery, producing a cleaned dataset of 84,643 U.S. observations for regression modeling.
- Trained and compared OLS, Ridge, and Lasso regression models in scikit-learn to predict daily maximum temperature, achieving test RMSE of 3.79°F and $R^2 = 0.969$ on nearly 17,000 held-out observations.

Daily Uber Trip Forecasting (NYC TLC) | R · SARIMA · ARMAX · forecast · astsa May 2026

- Fit a SARIMA(2,0,1)(2,1,0)[7] baseline via Box-Jenkins pipeline and auto.arima model search across 30+ candidate specifications on 365 days of NYC TLC HVFHV trip records, achieving a training MAPE of 3.97% and RMSE of 26,189.
- Extended to an ARMAX model with average fare, trip miles, and trip duration as external regressors, improving MAPE to 3.25% and RMSE to 23,130.

NLP Movie Recommendation System | R · TF-IDF · Cosine Similarity · tidytext May 2026

- Processed raw data across 4 merged files by parsing JSON strings, resolving missingness, and deduplicating 45,000+ records; tokenized 17M+ words and extracted bigrams to compute TF-IDF weights for cosine similarity-based recommendations.
- Implemented a hybrid ranking algorithm combining cosine similarity with weighted TMDb popularity and user ratings, validating improvement over pure similarity ranking via side-by-side model comparison.

Payment Fraud Classification (IEEE-CIS) | R · tidymodels · ggplot2 March 2026

- Built and tuned four classification models (Logistic Regression, Elastic Net, Random Forest, XGBoost) on 590K+ real-world credit card transactions under severe class imbalance (3.5% fraud rate).
- Achieved test ROC AUC of 0.942 and PR AUC of 0.888 with XGBoost, outperforming linear baselines by 13%; identified dominant fraud signals via feature importance analysis.

EXPERIENCE

Product Guide

July 2022 – September 2024

Arc'teryx

Livermore, CA

- Generated \$900,000+ in revenue across 3,000+ sales with a 12% conversion rate by analyzing client engagement to inform cross-sell strategy and product recommendations.

Business Owner

May 2021 – April 2024

Valley Soles

Tracy, CA

- Managed pricing, sourcing, and inventory for an independent resale business over 3 years, generating \$40,000 in profit by analyzing market demand across 1,000+ units (30–60% margins).